

Mixer

The mixer module, which can be built in two different versions, is designed for mixing any combination of control voltages, modulation signals, and audio signals. To this end, the circuit features an input for external signals (microphone, electric guitar, electric organ, etc.). Individual and master levels are adjustable, and an LED indicator warns of signal clipping. At the output, both an inverted and a non-inverted version of the summed signal are simultaneously available. Unless you happen to possess "splitter patch cords"—cables featuring multiple plugs at one end—you will find it quite difficult to execute certain control and modulation processes without a dedicated mixer. For instance, if you wish to modulate the PWM of a VCO using two LFOs simultaneously, this presents no problem whatsoever provided you have a mixer module at your disposal.

The Circuit

The circuit's functional operation is illustrated in Figure [1](#). Although the circuit design is quite simple, the following section provides a brief description of the function of each individual circuit block.

The operational amplifier (op-amp) stage, built around IC1, is sufficiently sensitive to amplify signals ranging from 10 mVpp to approximately 1.5 Vpp—a level consistent with the standard signal levels found in the FORMANT system. Consequently, the sensitivity of this amplifier stage is high enough to accommodate both guitar pickups and condenser microphones. The level of the input signal can be attenuated using potentiometer P1. Diodes D1 and D2 serve to limit excessive signal amplitudes.

The output signal from IC1—combined with up to four other input signals (for a maximum total of five), each of whose levels can also be individually attenuated—is fed into an inverting summing amplifier (IC2); the gain of this stage can be adjusted via potentiometer P4. The overall summed signal level can be attenuated using potentiometer P5.

IC3 is configured as a unity-gain inverter, thereby restoring the original signal polarity at its output. The outputs are decoupled via low-impedance drivers. The LED driver circuit, built around IC4, provides a "linearized" visual indication of the non-inverted output signal level and serves as a warning indicator for signal clipping (overload).

Construction and Calibration

The provided PCB layout and component placement diagram (Figure [2](#)) ensure a straightforward and trouble-free assembly process. Those who do not require a dedicated input for external signals may choose to omit the circuit block enclosed within the dashed frame around IC1 (as shown in Figure [1](#)). Pin 3 of IC1 and the cathode of D2 must be connected using

a wire link or stranded wire. For Input 3, a 3.5 mm jack socket should then be provided instead of a 6.35 mm type. Since the mounting hole for the Input 3 jack socket has a diameter of 9 mm, the best approach is to use a two-component adhesive (e.g., UHU-endfest 300) to bond a 1.5 mm thick aluminum plate—pre-drilled with a 6 mm hole—to the rear surface of the front panel to facilitate mounting of the 3.5 mm jack socket.

Alternatively, it is also possible to expand the unit to a total of 6 inputs by adding additional 470 nF capacitors, 10 k Ω potentiometers, and 22 k Ω resistors. Figure 3 presents suggested front panel layouts for both versions. The front panel designed for 6 inputs is twice as wide as standard "small" FORMANT front panels. Due to space constraints, only potentiometer types with 4 mm shafts can be used for the input level controls.

Calibration: P4 is used to set the gain of IC2. This allows the module to be adapted to specific requirements. P6 is adjusted such that a total output signal of 3 V_{pp} causes the LED to light up faintly, yet distinctly.

Application Notes

The mixer can be used to combine any control voltages, modulation signals, or audio signals. For instance, simultaneous modulation of VCOs, VCFs, etc., by multiple LFOs is easily achievable. Furthermore, the mixer can be utilized to integrate banks of VCOs or VCF/VCA modules into the signal chain. The mixer also proves highly useful when processing external signals within a FORMANT system, as it boosts the incoming signal to the standard operating level typically found in FORMANT modules. Thus, a simple guitar synthesizer can be constructed using the mixer in conjunction with several other FORMANT modules (see Figures 6 and 7). An old foot pedal from an electric sewing machine can serve effectively as a momentary switch. For more sophisticated guitar synthesizers, a significantly greater effort is required: a dedicated pickup for each string, 6 PTVCs (Pitch-to-Voltage Converters), 6 VCOs, etc. (see Figure 8).

Important Note!

To mix slow LFO signals or DC voltages, inputs 1 and 2 must be DC-coupled. To achieve this, capacitors C2 and C3 on the circuit board must be replaced with wire jumpers.

Figures

1

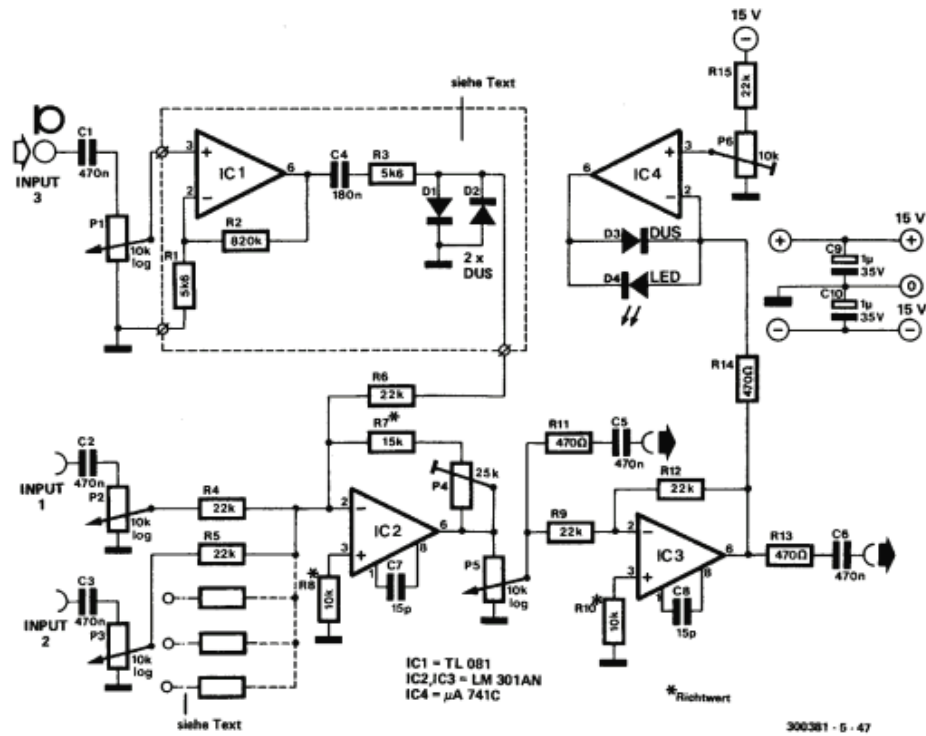


Figure 1. Schematic diagram of the universal mixer.

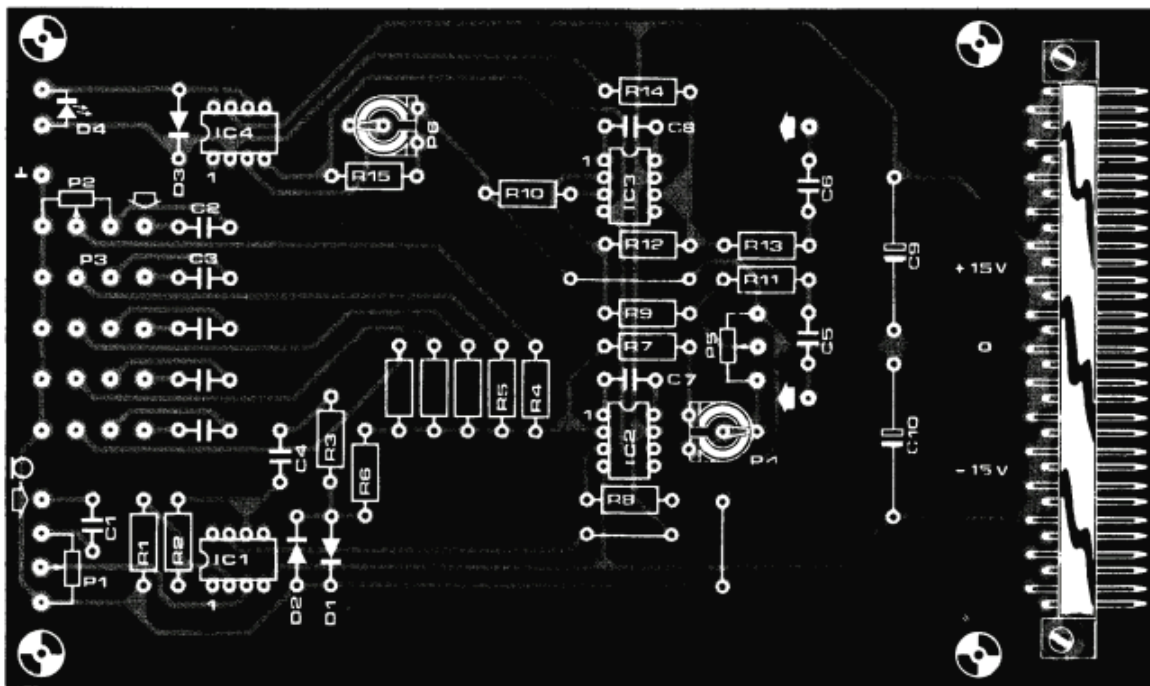
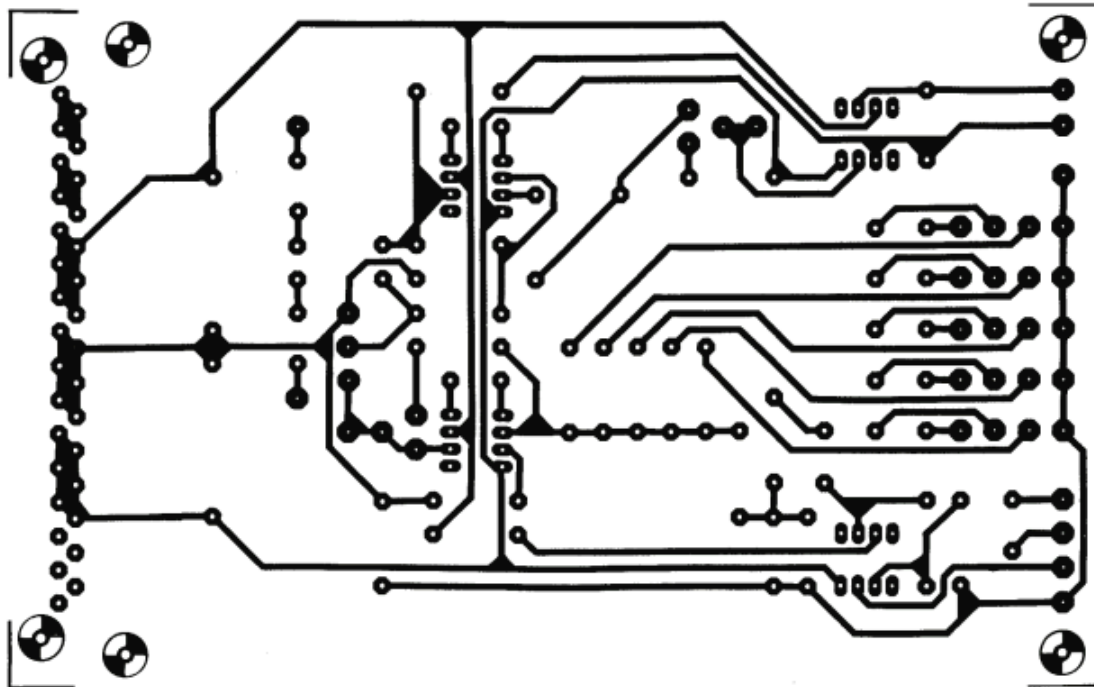


Figure 2. PCB layout and component placement plan for the mixer module. The circuit board is suitable for use in both the small and large versions. If the ES input is omitted, the framed section of the circuit is left out. In its place, a wire jumper is installed running from Pin 3 (IC1) to the cathode of D2.

3

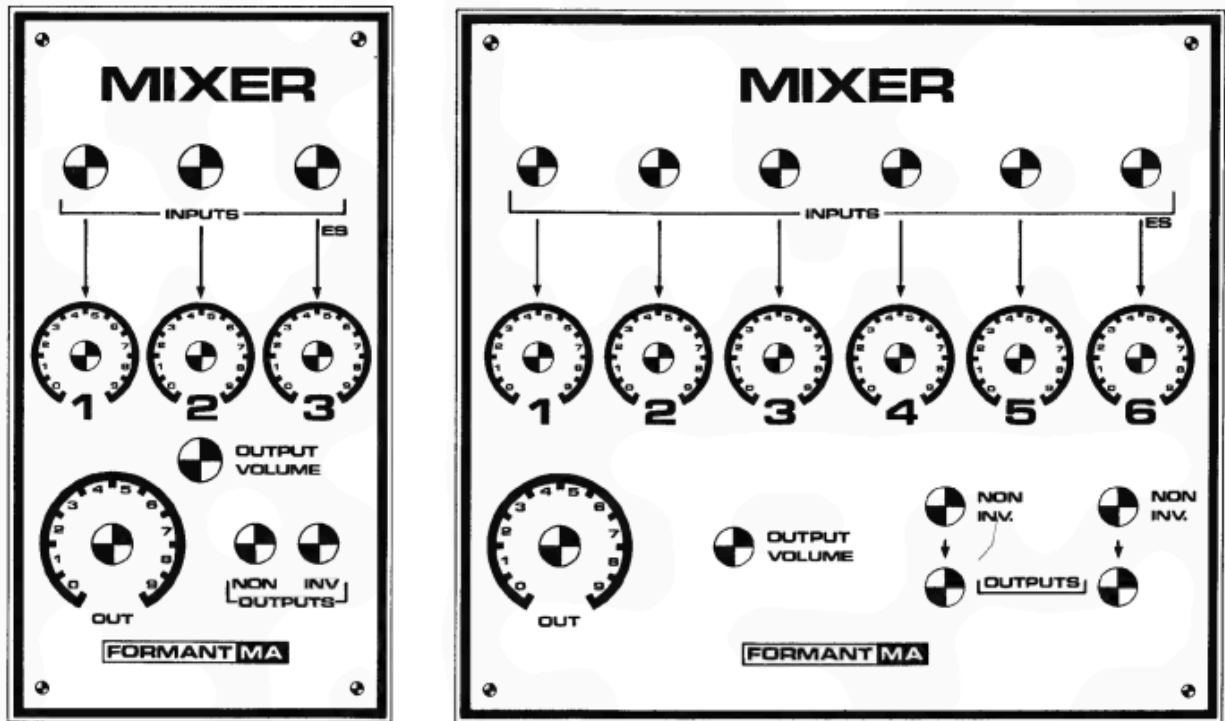


Figure 3. Suggested front panel layout for the mixer. Depending on the specific configuration, the small format may suffice. For a total of 6 inputs, however, the large format must be selected. The ES input is intended for external signal sources (i.e., those located outside the FORMANT unit).

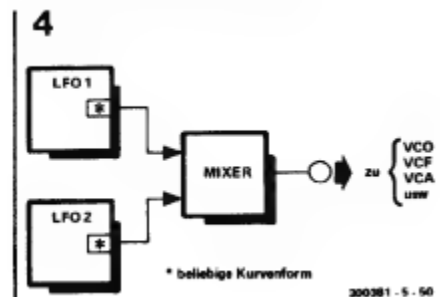


Figure 4. Basic configuration for multi-modulation using two LFOs.

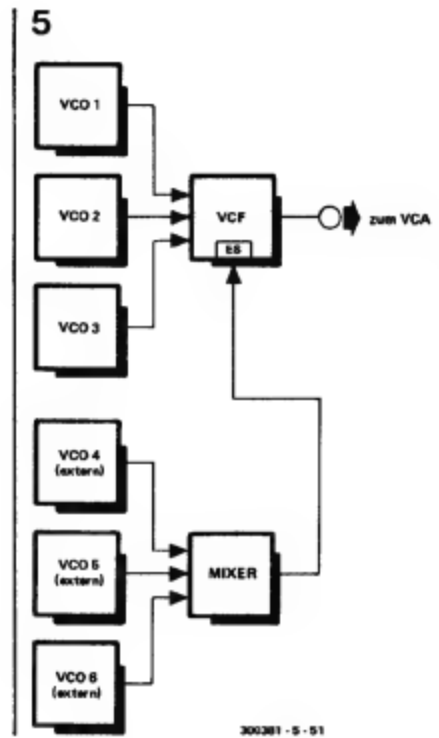


Figure 5. Integrating an external VCO bank (i.e., one housed in a separate enclosure) into the FORMANT's signal path.

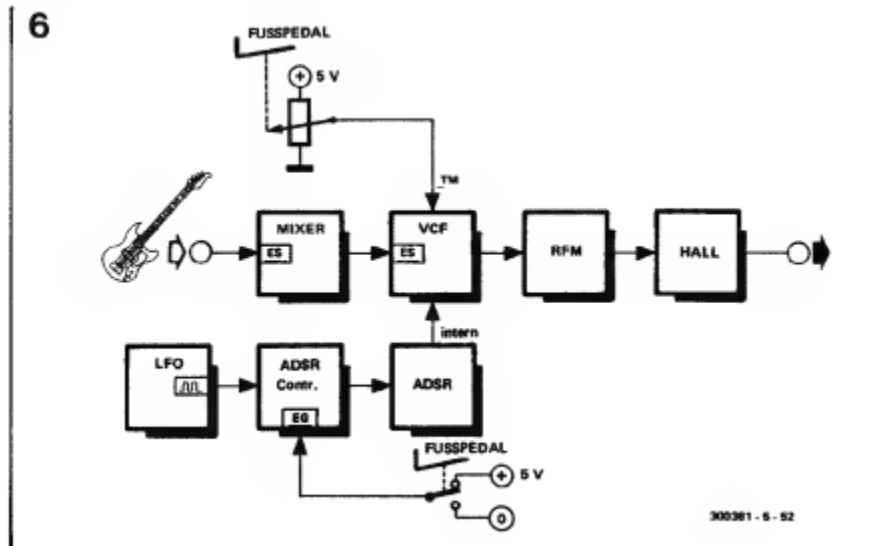
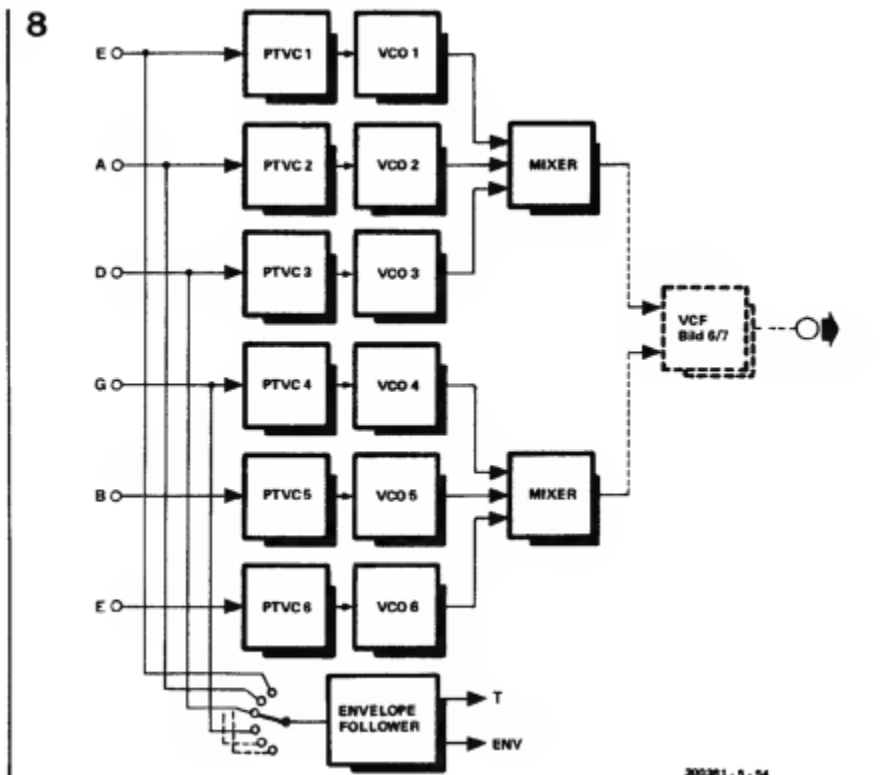
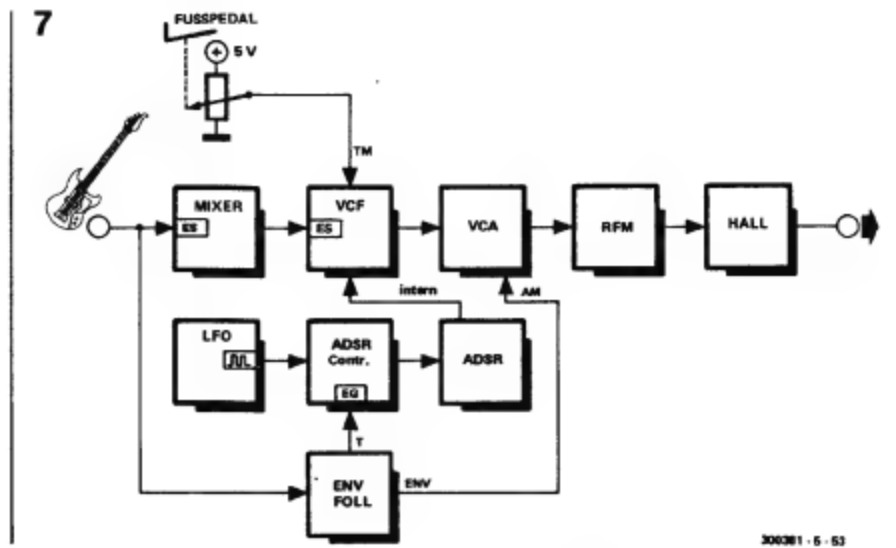


Figure 6. Basic configuration for a guitar or drum synthesizer, featuring sound shaping via a pedal (TM) and a foot-operated pushbutton (ADSR).



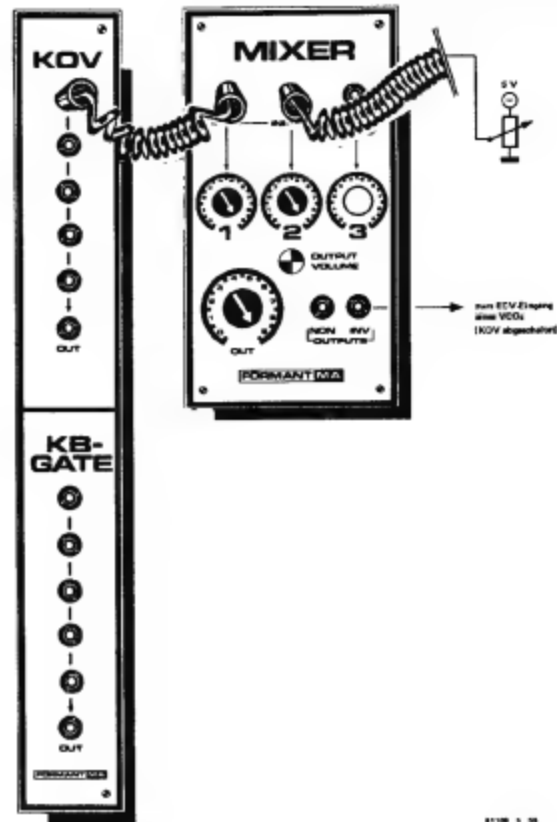


Figure 9. Inversion of the KOV (Keyboard Output Voltage). A typical example demonstrating a potential application for two expansion modules: The KOV is inverted within the mixer, causing notes at the upper end of the keyboard to sound lower in pitch than those at the lower end. The negative voltage supplied via the mixer's second input allows the full frequency range of the VCO to be utilized without the need to adjust its OCTAVES control. The setting of the potentiometer—which may also take the form of a foot pedal—is relatively non-critical.

10a



b

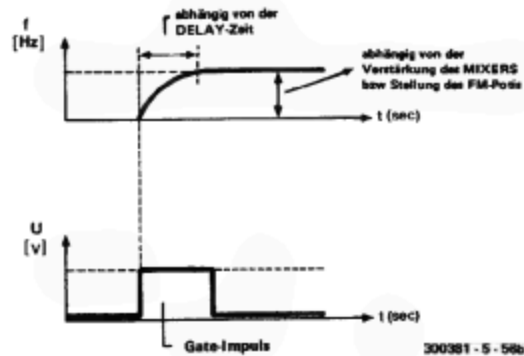


Figure 10. Using the mixer to generate a glide effect. As can be seen from the timing diagram in Figure 10b, the glide duration depends on the Decay time of the ADSR envelope, while the frequency offset depends on the mixer gain (or, alternatively, on the setting of the VCO's FM control). When the envelope runs automatically (switch set to "AD!"), the glide effect is independent of the duration of the Gate signal (or key press). Effects featuring longer Attack times also yield musically interesting results. Naturally, in addition to FM, both TM and AM can also be modulated.

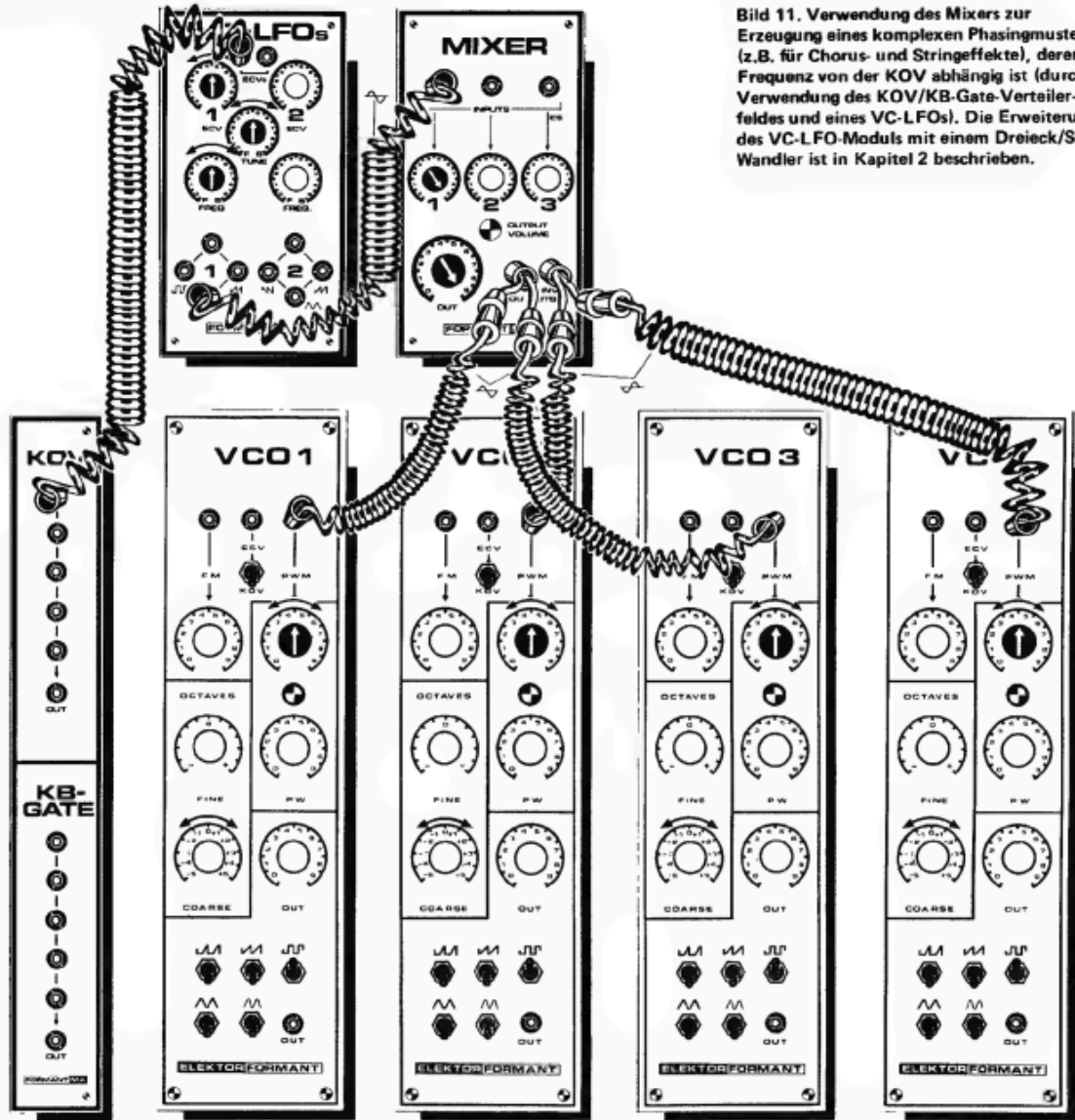


Bild 11. Verwendung des Mixers zur Erzeugung eines komplexen Phasingmusters (z.B. für Chorus- und Stringeffekte), deren Frequenz von der KOV abhängig ist (durch Verwendung des KOV/KB-Gate-Verteilerfeldes und eines VC-LFOs). Die Erweiterung des VC-LFO-Moduls mit einem Dreieck/Sinus-Wandler ist in Kapitel 2 beschrieben.

Figure 11. Using the mixer to generate a complex phasing pattern (e.g., for chorus and string effects), where the frequency is dependent on the KOV (achieved by utilizing the KOV/KB Gate distribution panel in conjunction with a VC-LFO). The expansion of the VC-LFO module to include a triangle-to-sine converter is described in Chapter 2.

Mixer Bill of Materials (for max. 3 inputs)

Resistors (Carbon Film, 5%):

- R1, R3 = 5.6k Ω
- R2 = 820k Ω
- R4, R5, R6, R9, R12, R15 = 22k Ω
- R7 = 15k Ω
- R8, R10 = 10k Ω
- R11, R13, R14 = 470 Ω

Potentiometers:

- P1, P2, P3, P5 = 10k Ω log. (logarithmic)
- P4 = 25k Ω (Trimmer)
- P6 = 10k Ω (Trimmer)

Capacitors:

- C1, C2, C3, C5, C6 = 470nF (MHM, MKS)
- C4 = 18nF (150nF + 33nF) (MHM, MKS)
- C7, C8 = 15pF (Ceramic)
- C9, C10 = 1 μ F/35V

Semiconductors:

- IC1 = TL081C
- IC2, IC3 = LM301AN
- IC4 = μ A741C (Mini-DIP)
- D1, D2, D3 = Diodes (e.g., 1N4148)
- D4 = LED

Miscellaneous:

- 1 x 31-pin header strip or solder pins
- 4 x 3.5mm jack sockets
- 1 x 6.35mm jack socket (see text)
- 1 x Control knob for 6mm shaft
- 3 x Control knobs for 4mm shaft